

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

MEDICAL : NURTURE PHASE-I & II

TEST # 03

DATE : 17-06-2018

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

Your Target is to secure Good Rank in AIIMS 2020

PART-I [PHYSICS]

- | | |
|---|--|
| <p>1. Which of the following is a scalar quantity ?
 (1) Work (2) Displacement
 (3) Velocity (4) Acceleration</p> <p>2. If $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$, then the angle between $\vec{A}$ and $\vec{B}$ is :
 (1) π (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$</p> <p>3. A physical quantity which has a direction :
 (1) must be a vector
 (2) may be a vector
 (3) may be both scalar and vector
 (4) none of the above</p> <p>4. Which of the following physical quantities are represented by axial vectors ?
 (1) Displacement (2) Force
 (3) Velocity (4) Torque</p> <p>5. Which of the following physical quantities are represented by polar vectors ?
 (1) Displacement
 (2) Angular velocity
 (3) Angular momentum
 (4) Torque</p> <p>6. The two vectors have magnitudes 3 and 5. If angle between them is 60° then the dot product of two vectors will be :
 (1) 7.5 (2) 6.5
 (3) 8.4 (4) 7.9</p> | <p>7. The minimum number of vectors of equal magnitude required to produce a zero resultant is:
 (1) 2 (2) 3
 (3) 4 (4) more than 4</p> <p>8. The minimum number of vectors of unequal magnitude required to produce a zero resultant is:
 (1) 2 (2) 3
 (3) 4 (4) more than 4</p> <p>9. What is the maximum number of components into which a vector can be split ?
 (1) 2 (2) 3
 (3) 4 (4) Infinite</p> <p>10. What is the maximum number of rectangular components into which a vector can be split in space ?
 (1) 2 (2) 3
 (3) 4 (4) Infinite</p> <p>11. The vector sum of the forces of 10 newton and 6 newton can be :
 (1) 2N (2) 8N (3) 18N (4) 20N</p> <p>12. Vectors sum of two forces of 10N and 6N cannot be :
 (1) 4N (2) 8N (3) 12N (4) 2N</p> <p>13. The vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} + \vec{B} = \vec{A} - \vec{B}$ then the angle between the two vectors $\vec{A}$ and $\vec{B}$ will be :
 (1) 0° (2) 60° (3) 90° (4) 180°</p> |
|---|--|

SPACE FOR ROUGH WORK

14. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A} + \vec{B} = \vec{C}$ and $A^2 + B^2 = C^2$. If θ is the angle between positive directions of $\vec{A}$ and $\vec{B}$ then mark the correct alternative :

(1) $\theta = 0^\circ$ (2) $\theta = \frac{\pi}{2}$ (3) $\theta = \frac{2\pi}{3}$ (4) $\theta = \pi$

15. If $\vec{A} = \vec{B} + \vec{C}$ and the magnitudes of $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 5, 4 and 3 units respectively, the angle between $\vec{A}$ and $\vec{C}$ is :

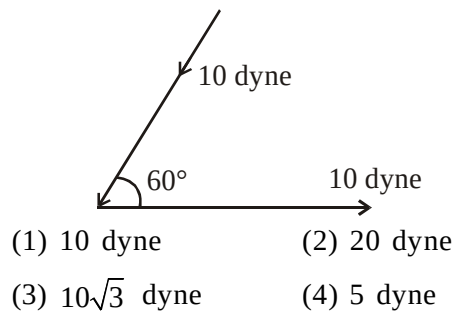
(1) $\cos^{-1}\left(\frac{3}{5}\right)$ (2) $\cos^{-1}\left(\frac{4}{5}\right)$

(3) $\frac{\pi}{2}$ (4) $\sin^{-1}\left(\frac{3}{4}\right)$

16. If $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$, then angle θ between vectors $\vec{A}$ and $\vec{B}$ is :

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

17. Two forces each numerically equal to 10 dyne are acting as shown in the adjoining figure, then the resultant is :



18. A force vectors applied on a mass is represented as $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ and the mass accelerated with 1 m/s^2 what will be the mass of the body?

(1) $10\sqrt{2}$ kg (2) $2\sqrt{10}$ kg
(3) 10 kg (4) 20 kg

19. Let $\vec{A} = \hat{i}A \cos \theta + \hat{j}A \sin \theta$ be any vector. Another vector $\vec{B}$ which is normal to $\vec{A}$ is :

(1) $\hat{i}B \cos \theta + \hat{j}B \sin \theta$ (2) $\hat{i}B \sin \theta + \hat{j}B \cos \theta$
(3) $\hat{i}B \sin \theta - \hat{j}B \cos \theta$ (4) $\hat{i}A \cos \theta - \hat{j}A \sin \theta$

20. The angle between two vectors $2\hat{i} + 3\hat{j} + \hat{k}$ and $-3\hat{i} + 6\hat{k}$ is :

(1) 0° (2) 45° (3) 60° (4) 90°

21. If a unit vector is represented by $0.5\hat{i} + 0.8\hat{j} + c\hat{k}$, then the value of 'c' is :

(1) 1 (2) $\sqrt{0.11}$ (3) $\sqrt{0.01}$ (4) $\sqrt{0.39}$

22. If $\vec{A} + \vec{B} = \vec{C}$ and $A = B = C$, then what should be the angle between $\vec{A}$ and $\vec{B}$?

(1) 0 (2) $\frac{\pi}{3}$ (3) $\frac{2\pi}{3}$ (4) π

23. If $\vec{A} \times \vec{B} = 0$ and $\vec{A} \cdot \vec{B} = -AB$, then angle between $\vec{A}$ and $\vec{B}$ is ?

(1) zero (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

SPACE FOR ROUGH WORK

24. The diagonals of a parallelogram are represented by vectors $\vec{P} = 5\hat{i} - 4\hat{j} + 3\hat{k}$ and $\vec{Q} = 3\hat{i} + 2\hat{j} - \hat{k}$. Then the area of the parallelogram is :
- (1) $\sqrt{171}$ unit (2) $\sqrt{72}$ unit
(3) 171 unit (4) 72 unit
25. The vector $\vec{p}$ makes 120° with the x-axis and the vector $\vec{Q}$ makes 30° with the y-axis. What is their resultant ?
- (1) $\vec{P} + \vec{Q}$ (2) $\vec{P} - \vec{Q}$
(3) $\sqrt{P^2 + Q^2}$ (4) $\sqrt{P^2 - Q^2}$
26. A force of $(3\hat{i} + 4\hat{j})$ newton acts on a body and displaces it by $(3\hat{i} + 4\hat{j})$ metres the work done by the force is :
- (1) 10 J (2) 12 J (3) 16 J (4) 25 J
27. Two equal vectors have a resultant equal to either of them. The angle between them is :
- (1) 60° (2) 90°
(3) 100° (4) 120°
28. If the vectors $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$ and $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$ are perpendicular to each other, then the positive value of a is :
- (1) 3 (2) 2 (3) 1 (4) 0
29. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then the value of α is :
- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) 1 (4) -1
30. If $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is:
- (1) $(A^2 + B^2 + AB)^{1/2}$
(2) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
(3) $(A + B)$
(4) $(A^2 + B^2 + \sqrt{3}AB)^{1/2}$
31. If $y = \frac{x^2}{(x+1)}$ then $\frac{dy}{dx}$ is :
- (1) $\frac{x(3x+2)}{(x+1)^2}$ (2) $\frac{-x(x+2)}{(x+1)^2}$
(3) $\frac{x(x+2)}{(x+1)^2}$ (4) $\frac{x(x+2)}{(x+1)}$
32. If $y = \log_e x + \sin x + e^x$ then $\frac{dy}{dx}$ is :
- (1) $\frac{1}{x} + \sin x + e^x$ (2) $\frac{1}{x} - \cos x + e^x$
(3) $\frac{1}{x} + \cos x + e^x$ (4) $\frac{1}{x} - \sin x$

SPACE FOR ROUGH WORK

33. $\frac{d}{dx}(\sin 120^\circ) = \dots\dots\dots$

- (1) $\cos 120^\circ$
- (2) $120 \cos 120$
- (3) 0
- (4) None of these

34. If $y = x^3 \cos x$ then $\frac{dy}{dx} = \dots\dots\dots$:

- (1) $x^2(3\cos x - x\sin x)$
- (2) $x^2(3\cos x + x\sin x)$
- (3) $3x^2\cos x + x^3\sin x$
- (4) None of these

35. If $y = \sin x + \cos x$ then $\frac{d^2y}{dx^2}$ is :

- (1) $\sin x - \cos x$
- (2) $\cos x - \sin x$
- (3) $-(\sin x + \cos x)$
- (4) None of these

36. The value of integral $\int_2^4 \frac{dx}{x}$ is :

- (1) $3 \log_e 2$
- (2) $\log_e 2$
- (3) $\log_e 4$
- (4) $2 \log_e 8$

37. If $2\cos 2\theta = \sqrt{3}$, then $\theta = ?$

- (1) 30°
- (2) 90°
- (3) 45°
- (4) 15°

38. $r = x^3 + y^3 + z^3$ find $\frac{\delta r}{\delta y}$:

- (1) $3y^2$
- (2) $x^3 + 3y^2 + z^3$
- (3) $3x^2 + 3y^2 + 3z^2$
- (4) zero

39. $\int_0^{\pi/4} \sec x \tan x \, dx$:

- (1) $\sqrt{1} - 2$
- (2) $\sqrt{2} - 2$
- (3) $-\frac{3}{14}$
- (4) $\sqrt{2} - 1$

40. Range of the $\sin x$ function is :

- (1) $[-1, 1]$
- (2) $[0, +\infty]$
- (3) $(-\infty, +\infty)$
- (4) $[-2, 2]$

SPACE FOR ROUGH WORK

PART-II [CHEMISTRY]

- | | |
|--|--|
| <p>41. Atomic number of an element represents.</p> <p>(1) number of neutrons in the nucleus
 (2) atomic mass of an element
 (3) valency of an element
 (4) number of protons in the nucleus</p> <p>42. Rutherford's scattering experiment is related to the size of the :</p> <p>(1) Nucleus (2) Atom
 (3) Electron (4) neutron</p> <p>43. Atoms may be regarded as comprising of protons, neutrons and electrons. If the mass attributed by electrons was doubled and that attributed by neutrons was halved, the atomic mass of ^{12}C would be :</p> <p>(1) approximately the same
 (2) doubled
 (3) reduced approx. 25 %
 (4) approx halved</p> <p>44. Which of the following atoms contains the least number of neutrons ?</p> <p>(1) $^{235}_{92}\text{U}$ (2) $^{238}_{92}\text{U}$
 (3) $^{239}_{93}\text{U}$ (4) $^{240}_{93}\text{U}$</p> <p>45. Increasing order (lowest first) for the values of e/m (charge/mass) for electron (e), proton (p), neutron (n) and α-particle (α) is :</p> <p>(1) e, p, n, α (2) n, p, e, α
 (3) n, p, α, e (4) n, α, p, e</p> | <p>46. Isotopes of an element have.</p> <p>(1) different chemical and physical properties
 (2) similar chemical and physical properties
 (3) similar chemical and different physical properties
 (4) similar physical and different chemical properties</p> <p>47. Which of the following pairs represents isobars?</p> <p>(1) ^3_2He and ^4_2He
 (2) $^{24}_{12}\text{Mg}$ and $^{25}_{12}\text{Mg}$
 (3) $^{40}_{19}\text{K}$ and $^{40}_{20}\text{Ca}$
 (4) $^{40}_{19}\text{K}$ and $^{39}_{19}\text{K}$</p> <p>48. The ratio of energy of radiations of wavelengths 2000 Å and 4000 Å is :</p> <p>(1) 2 (2) 4
 (3) 1/2 (4) 1/4</p> <p>49. The ratio of the volume of the atom and the volume of the nucleus is :</p> <p>(1) 10^{10} (2) 10^{12}
 (3) 10^{15} (4) 10^{20}</p> <p>50. A 600 W mercury lamp emits monochromatic radiation of wavelength 331.3 nm. How many photons are emitted from the lamp per second? ($h=6.626 \times 10^{-34}$ Js, velocity of light=$3 \times 10^8 \text{ms}^{-1}$)</p> <p>(1) 1×10^{19} (2) 1×10^{20}
 (3) 1×10^{21} (4) 1×10^{23}</p> |
|--|--|

SPACE FOR ROUGH WORK

51. Which of the following statements does not form a part of Bohr's model of hydrogen atom?
 (1) Energy of the electrons in the orbit is quantised
 (2) The electron in the orbit nearest to the nucleus has the lowest energy
 (3) Electrons revolve in different orbits around the nucleus
 (4) The position and velocity of the electrons in the orbit cannot be determined simultaneously
52. The radius of the first orbit of H-atom is r . Then the radius of the first orbit of Li^{2+} will be :
 (1) $\frac{r}{9}$ (2) $\frac{r}{3}$
 (3) $3r$ (4) $9r$
53. The distance between 3rd and 2nd orbits in the hydrogen atom is :
 (1) $2.646 \times 10^{-8} \text{ cm}$ (2) $2.116 \times 10^{-8} \text{ cm}$
 (3) $0.058 \times 10^{-8} \text{ cm}$ (4) $0.529 \times 10^{-8} \text{ cm}$
54. According to Bohr theory, the angular momentum for an electron of 5th orbit is :
 (1) $\frac{5h}{\pi}$ (2) $\frac{2.5h}{\pi}$ (3) $\frac{5\pi}{h}$ (4) $\frac{2.5\pi}{h}$
55. Which of the following electron transitions in hydrogen atom is associated with largest amount of energy ?
 (1) from $n = 1$ to $n = 2$ (2) from $n = 2$ to $n = 3$
 (3) from $n = \infty$ to $n = 1$ (4) from $n = 3$ to $n = 5$
56. The ratio of the difference between 1st to 2nd Bohr orbits energy to that between 2nd and 3rd orbits energy is :
 (1) $1/2$ (2) $1/3$ (3) $27/5$ (4) $5/27$
57. Which of the following transitions of an electron in hydrogen atom emits radiation of the lowest wavelength ?
 (1) $n_2 = \infty$ to $n_1 = 2$ (2) $n_2 = 4$ to $n_1 = 3$
 (3) $n_2 = 2$ to $n_1 = 1$ (4) $n_2 = 5$ to $n_1 = 3$
58. The spectral lines corresponding to the radiation emitted by an electron jumping from 6th, 5th and 4th orbits to second orbit belong to.
 (1) Lyman series (2) Balmer series
 (3) Paschen series (4) Pfund series
59. The speed of the electron in the 1st orbit of the hydrogen atom in the ground state is (c is the velocity of light).
 (1) $\frac{c}{1.37}$ (2) $\frac{c}{1370}$ (3) $\frac{c}{13.7}$ (4) $\frac{c}{137}$
60. A 200 g cricket ball is thrown with a speed of $3.0 \times 10^3 \text{ cm sec}^{-1}$. What will be its de Broglie's wavelength ? ($h = 6.6 \times 10^{-27} \text{ g cm}^2 \text{ sec}^{-1}$)
 (1) $1.1 \times 10^{-32} \text{ cm}$ (2) $2.2 \times 10^{-32} \text{ cm}$
 (3) $0.55 \times 10^{-32} \text{ cm}$ (4) $11.0 \times 10^{-32} \text{ cm}$
61. Which of the following expressions gives the de Broglie relationship ?
 (1) $p = \frac{h}{mv}$ (2) $\lambda = \frac{h}{mv}$
 (3) $\lambda = \frac{h}{mp}$ (4) $\lambda m = \frac{h}{p}$

SPACE FOR ROUGH WORK

62. The principal quantum number of an atom is related to the :
- size of the orbital
 - orbital angular momentum
 - spin angular momentum
 - orientation of the orbital in space
63. The quantum number not obtained from the Schrodinger's wave equation is :
- n
 - l
 - m
 - s
64. If the ionization energy for hydrogen atom is 13.6 eV, then the ionization energy for He^+ ion should be :
- 13.6 eV
 - 6.8 eV
 - 54.4 eV
 - 72.2 eV
65. Which of the following represents the correct set of four quantum numbers of 4d-electron ?
- 4, 3, 2, $+1/2$
 - 4, 2, 1, 0
 - 4, 3, -2 , $+1/2$
 - 4, 2, 1, $-1/2$
66. Values of magnetic quantum number for an electron of M-shell can be.
- 0, 1, 2
 - -2 , -1 , 0, $+1$, $+2$
 - 0, 1, 2, 3
 - -1 , 0, $+1$
67. In hydrogen atom, the electron is at a distance of 4.768 Å from the nucleus. The angular momentum of the electron is.
- $\frac{3h}{2\pi}$
 - $\frac{h}{2\pi}$
 - $\frac{h}{\pi}$
 - $\frac{3h}{\pi}$
68. In Bohr's model, if the atomic radius of the first orbit r_1 , then radius of fourth orbit will be.
- $4r_1$
 - $6r_1$
 - $16r_1$
 - $\frac{r_1}{16}$
69. α -particles are represented by
- lithium atoms
 - helium nuclei
 - hydrogen nucleus
 - none of the above
70. A certain radio station broadcasts on a frequency of 980 kHz (kilohertz). What is the wavelength of electromagnetic radiation broadcast by the radio station?
- 306 m
 - 3.06 m
 - 30.6 m
 - 3060 m
71. If the radius of third Bohr orbit be a_0 , then the radius of the first orbit would be
- $3 \times a_0$
 - $6 \times a_0$
 - $9 \times a_0$
 - $1/9 \times a_0$

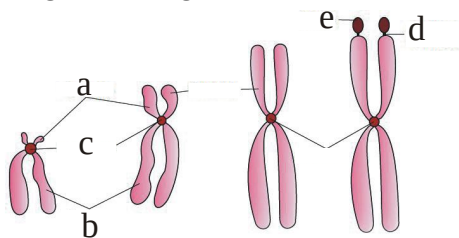
SPACE FOR ROUGH WORK

- | | |
|--|---|
| <p>72. The ratio of ionization energy of H and Be^{+3} is
 (1) 1 : 1 (2) 1 : 3
 (3) 1 : 9 (4) 1 : 16</p> <p>73. In two H atoms X and Y the electrons move around the nucleus in circular orbits of radius r and $4r$ respectively. The ratio of the times taken by them to complete one revolution is
 (1) 1 : 4 (2) 1 : 2
 (3) 1 : 8 (4) 2 : 1</p> <p>74. The ratio of the radii of first three Bohr orbits is
 (1) 1 : 0.5 : 0.33
 (2) 1 : 2 : 3
 (3) 1 : 4 : 9
 (4) 1 : 8 : 27</p> <p>75. If threshold wavelength (λ_0) for ejection of electron from metal is 330 nm, then work function for the photoelectric emission is -
 (1) 1.2×10^{-18} J
 (2) 1.2×10^{-20} J
 (3) 6×10^{-19} J
 (4) 6×10^{-12} J</p> <p>76. The kinetic energy of the electron emitted when light of frequency $3.5 \times 10^{15} \text{ Hz}$ falls on a metal surface having threshold frequency $1.5 \times 10^{15} \text{ Hz}$ is ($h = 6.6 \times 10^{-34} \text{ Js}$)
 (1) $1.32 \times 10^{-18} \text{ J}$ (2) $3.3 \times 10^{-18} \text{ J}$
 (3) $6.6 \times 10^{-19} \text{ J}$ (4) $1.98 \times 10^{-19} \text{ J}$</p> | <p>77. To which electronic transition between Bohr orbits in hydrogen, the second line in the Balmer series belongs ?
 (1) $3 \rightarrow 2$
 (2) $4 \rightarrow 2$
 (3) $5 \rightarrow 2$
 (4) $6 \rightarrow 2$</p> <p>78. Electromagnetic radiation with maximum wavelength is-
 (1) ultra violet
 (2) radio wave
 (3) X-ray
 (4) infra-red</p> <p>79. The nuclides $^{40}_{18}\text{Ar}$ and $^{41}_{19}\text{K}$ are -
 (1) isotopes
 (2) isobars
 (3) isotones
 (4) isoelectronic</p> <p>80. Calculate the number of quanta of radiations of frequency $4.67 \times 10^{13} \text{ s}^{-1}$ that must be absorbed in order to melt 5g of ice (The energy required to melt 1 g of ice is 333 J)
 (1) 3.36×10^{18}
 (2) 17
 (3) 5.38×10^{22}
 (4) 30.91×10^{-21}</p> |
|--|---|

SPACE FOR ROUGH WORK

PART-III [BIOLOGY]

- 81.** Which statement is not appropriate?
 (1) Centrioles are surrounded by amorphous pericentriolar bodies
 (2) Outer and inner nuclear membranes fuse to form nuclear pores at a number of places
 (3) Numerous nucleoli present in cells actively carrying out protein synthesis
 (4) Chromatin consist of DNA and RNA
- 82.** Which of the following pair is correct?
 (1) histone - acidic amino acids
 (2) non histone - basic amino acids
 (3) centromeres - first site of constriction
 (4) kinetochore - cylindrical structure
- 83.** Highest numbers of chloroplast are present in
 (1) stem (2) leaf
 (3) mesophyll cells (4) cortical cells
- 84.** Recognize the figure and find the correct match

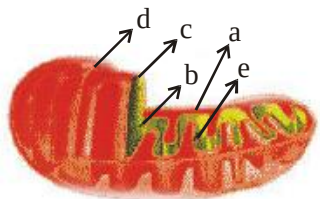


- (1) a-p arm, b-long arm, c-primary constriction, d-SAT
 (2) a-short arm, b-q arm, c-centromere, d-secondary constriction
 (3) a-q arm, b-p arm, c-centromere, d-satellite
 (4) a-p arm, b-q arm, c-primary constriction, e-secondary constriction

- 85.** Read the following statements. How many statements are correct ?
 (a) Basal granules have cart wheel like organization
 (b) ER helps in synthesis of lipoproteins and glycogen
 (c) RuBisCO is most abundant protein in biosphere
 (d) The number of chloroplast varies from 1 per cell in euglena to 20-40 per cell in the mesophyll.
 (e) Enzymes for ETS are embedded in inner membrane of mitochondria
 (f) Pili and fimbriae are found in some bacteria. Pili are shorter in length than fimbriae
- (1) three (2) four
 (3) five (4) six
- 86.** _____ are present in germinating seeds and concerned with oxidation of fats.
 (1) Peroxisomes (2) Glyoxysomes
 (3) Sphaerosomes (4) Lomasomes
- 87.** Which stain is not used for staining chromosomes?
 (1) Methylene blue
 (2) Eosin
 (3) Basic fuchsin
 (4) Acetocarmine

SPACE FOR ROUGH WORK

88. Identify the figure and find the correct match



- (1) a-inner membrane , b-crista , c-outer membrane , d-perimitochondrial space, e-matrix
- (2) a-Intermembrane space , b-matrix , c-outer membrane, d-inner membrane, e-crista
- (3) a -perimitochondrial space , b-matrix, c-inner membrane , d-outer membrane , e-crista
- (4) a-crista, b-matrix , c-inner membrane, d-outer membrane, e-perimitochondrial space

89. Read the following statements carefully. How many statements are incorrect?

- (a) Cell hypothesis was proposed by T. Schwann
 - (b) Size of bacteria - 2 – 5 micron
 - (c) Mesosome is formed by infoldings of plasma membrane in bacteria and some protists
 - (d) Plasmid is small linear DNA present outside the genomic DNA
 - (e) Gas vacuoles are found in purple and green photosynthetic bacteria
 - (f) Membrane proteins are classified on the basis of nature of amino acids
- (1) One (2) Two (3) Three (4) Four

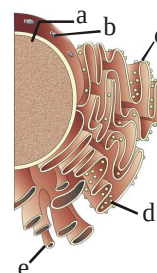
90. Concentration of ions is higher in sap vacuole than cytoplasm because

- (1) ions diffuse from cytoplasm into vacuole
- (2) tonoplast is more permeable for ions than other materials
- (3) tonoplast has active transport system that pump ions from cytoplasm into sap vacuole
- (4) interior side of tonoplast is impermeable to ions

91. Golgi body

- (1) known as lytic vacuole in plants
- (2) site of phosphorylation
- (3) occur in prokaryotes and eukaryotes
- (4) modifies proteins

92. Identify the figure and find the correct match



- (1) a -nucleus, b -nucleopore , c -RER, d -ribosomes, e-SER
- (2) a -nucleus, b - nucleopore, c - SER, d - ribosomes, e-RER
- (3) a-nucleolus, b -nucleopore, c -RER, d -ribosomes, e-SER
- (4) a -nucleolus, b -nucleopore , c -SER, d -ribosomes, e-RER

SPACE FOR ROUGH WORK

93. Match the column

	Organism		Diploid number of chromosomes
(a)	Haplopappus gracilis	(i)	14
(b)	Pisum sativum	(ii)	1262
(c)	Aulocantha	(iii)	4
(d)	Ophioglossum reticulatum	(iv)	1600

- (1) a-i, b-ii, c-ii, d-iv (2) a-i, b-ii, c-iv, d-ii
 (3) a-iii, b-i, c-ii, d-iv (4) a-iii, b-i, c-iv, d-ii

94. Chromatophores are found in some bacteria. They participate in

- (1) locomotion (2) respiration
 (3) photosynthesis (4) reproduction

95. Single membrane bound organelle is

- (1) ER (2) Golgi body
 (3) Lysosome (4) All

96. Polysomes are found in

- (1) prokaryotes only (2) eukaryotes only
 (3) both 1 and 2 (4) none

97. How many statements are not correct?

- (a) Telomeres are non sticky ends of chromosomes
 (b) Each chromatid has a pair of chromonema
 (c) A chromonema is coiled DNA thread
 (d) Submetacentric chromosomes appears j shaped during anaphase
 (e) DNA is the chemical basis of heredity
 (f) Genes are hereditary vehicles
 (1) one (2) two (3) three (4) four

98. Synthesis of r-RNA occurs in

- (1) nucleus (2) nucleolus
 (3) cytoplasm (4) ribosome

99. Which structures are made up of microtubules?

- (1) Centrosome, nucleosome, centrioles
 (2) Cilium, centriole, spindle fibre
 (3) Centriole, spindle fibre, chromatin
 (4) Cilium, flagellum, ergastic substance

100. Which of the following statement is not correct?

- (1) The thylakoid membrane contains photosynthetic pigments such as chlorophyll, carotenes and xanthophylls
 (2) Lysosomes help in remodeling of cartilages and bones.
 (3) Microfilaments are present in muscle fibers and associated with myosin protein
 (4) None of these

101. The four elements called "**Big-four**" which makes up 95% of all elements found in a living system are

- (1) C, H, O, N (2) C, H, O, P
 (3) C, H, O, S (4) C, N, O, P

102. Macromolecules are

- (1) nucleic acids, proteins and polysaccharides
 (2) nucleic acids and monosaccharides
 (3) amino acids and polysaccharides
 (4) amino acids, lipids and nucleotides

103. Glucose is

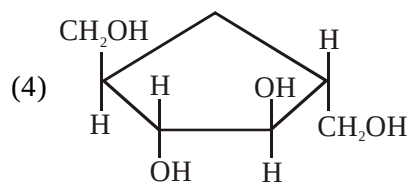
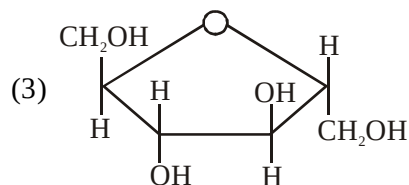
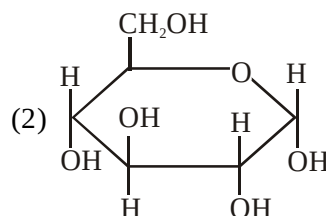
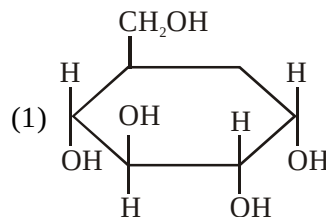
- (1) aldose hexose sugar
 (2) ketose hexose sugar
 (3) pyranose pentose sugar
 (4) furanose pentose sugar

SPACE FOR ROUGH WORK

- 104.** Reducing sugars are
 (1) glucose, fructose, galactose, maltose and lactose
 (2) glucose, sucrose and cellulose
 (3) lactose, starch, glycogen and trehalose
 (4) all of the above
- 105.** Reducing sugars like glucose in Fehling solution reduce
 (1) Fe^{++} to Fe^{+++} (2) Cu^{++} to Cu^{+}
 (3) Hg^{++} to Hg^{+} (4) Cu^{+} to Cu^{++}
- 106.** If deoxyribose sugar is supplemented with oxygen at second carbon atom, which one of these is formed ?
 (1) Erythrose (2) Heptose
 (3) Ribulose (4) Ribose
- 107.** The sweetest sugar is
 (1) fructose (2) glucose
 (3) sucrose (4) monellin
- 108.** General formula for disaccharide is
 (1) $\text{C}_n\text{H}_{2n+2}$ (2) $\text{C}_n(\text{H}_2\text{O})_{n+1}$
 (3) $\text{C}_n(\text{H}_2\text{O})_{n-1}$ (4) $\text{C}_{12}\text{H}_{22}\text{O}_{12}$
- 109.** The reagent used to detect sugar in the urine is:
 (1) Ninhydrin solution (2) Benzene
 (3) Benedict's solution (4) All of the above
- 110.** Lactose is a disaccharide of
 (1) glucose only
 (2) glucose and fructose
 (3) glucose and galactose
 (4) all of the above

- 111.** Prior to absorption, grape sugar is hydrolyzed by the enzyme
 (1) lactase (2) maltase
 (3) sucrose (4) none of these

- 112.** Choose the correct molecule for glucose.



SPACE FOR ROUGH WORK

113. Which of the following are all disaccharides ?
 (1) Maltose, Sucrose, Lactose
 (2) Maltose, Lactose, Mannose
 (3) Glycogen, Lactose, Sucrose
 (4) All of the above
114. Invert sugar is mixture of
 (1) maltose and fructose
 (2) glucose and galactose
 (3) glucose and fructose
 (4) all of the above
115. A solution of d-glucose in water rotates the plane polarised light
 (1) towards right (2) towards left
 (3) towards either side (4) none of the above
116. α and β -glucose differ in the orientation of $-OH$ group around
 (1) C_3 (2) C_1
 (3) C_5 (4) C_2
117. Monosaccharide found in nucleoside is
 (1) pentose (2) tetrose
 (3) erythrose (4) hexose
118. In ATP sugar is
 (1) ribose (2) deoxyribose
 (3) glucose (4) trioses
119. Non reducing sugars have
 (1) free $-CHO$ group and free $>CO$ group
 (2) neither free $>CO$ nor free $-CHO$ group
 (3) free $-CHO$ and bound $>CO$ group
 (4) free $>CO$ group and bound $-CHO$ group

120. Which should be given to an athlete for instant energy ?
 (1) Carbohydrates (2) Proteins
 (3) Fats (4) Vitamins

Direction for Assertion & Reason questions. These questions consist of two statements each, printed as assertion and reason. While answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the assertion.
 (2) If both Assertion & Reason are true but reason is not a correct explanation of the assertion.
 (3) If Assertion is true but the Reason is false
 (4) If both Assertion & Reason are false

121. **Assertion** :- Vector $(\hat{i} + \hat{j} + \hat{k})$ is perpendicular to $(\hat{i} - 2\hat{j} + \hat{k})$.

Reason :- Two non-zero vectors are perpendicular if their dot product is zero.

122. **Assertion** :- A physical quantity can be a vector, if magnitude as well as direction is associated with it.

Reason :- A physical quantity can be a scalar quantity, if it is associated with magnitude only

123. **Assertion** :- Distance is a scalar quantity.

Reason :- Distance is the length of path traversed

SPACE FOR ROUGH WORK

124. Assertion :- The dot product of one vector with another vector may be a scalar or a vector.

Reason :- If the product of two vectors is a vector quantity, then product is called a dot product

125. Assertion :- The angle between the two vectors

$$(\hat{i} + \hat{j}) \text{ and } (\hat{j} + \hat{k}) \text{ is } \frac{\pi}{3} \text{ radian.}$$

Reason :- Angle between two vectors $\vec{A}$ and

$$\vec{B} \text{ is given by } \theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$$

126. Assertion :- A vector can have zero magnitude if one of its rectangular components is not zero.

Reason :- Scalar product of 2 vectors cannot be a negative quantity.

127. Assertion :- The direction of a zero (null) vector is indeterminate.

Reason :- We can have $\vec{A} \times \vec{B} = \vec{A} \cdot \vec{B}$ with $\vec{A} \neq \vec{O}$ and $\vec{B} \neq \vec{O}$

128. Assertion :- A vector quantity is a quantity that has both magnitude and a direction and obeys the triangle law of addition and equivalently the parallelogram law of addition.

Reason :- The magnitude of the resultant vector of two given vectors can never be less than the magnitude of any of the given vector.

129. Assertion :- If the initial and final positions coincide, the displacement is a null vector.

Reason :- A physical quantity cannot be called a vector, if its magnitude is zero.

130. Assertion :- We can find angle between two vectors by using the relation.

$$\theta = \sin^{-1} \left[\frac{|\vec{A} \times \vec{B}|}{AB} \right]$$

Reason :- $\frac{|\vec{A} \times \vec{B}|}{AB}$ is always positive

131. Assertion :- $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{A})$ is $-A^2 B^2 \sin^2 \theta$.

Here θ is the angle between $\vec{A}$ and $\vec{B}$.

Reason :- $(\vec{A} \times \vec{B})$ and $(\vec{B} \times \vec{A})$ are two anti-parallel vectors provided A and B are neither parallel nor anti-parallel

132. Assertion :- If $|\vec{A}| = |\vec{B}|$, then $(\vec{A} + \vec{B}), (\vec{A} - \vec{B})$

and $(\vec{A} \times \vec{B})$ are three mutually perpendicular vectors.

Reason :- Dot product of a null vector with any other vector is always zero.

SPACE FOR ROUGH WORK

133. Assertion :- $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$.

Reason :- $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$ lie in the plane containing $\vec{A}$ and $\vec{B}$ while $\vec{A} \times \vec{B}$ lies perpendicular to the plane containing $\vec{A}$ and $\vec{B}$.

134. Assertion :- If angle between $\vec{a}$ and $\vec{b}$ is 30° .

Then angle between $2\vec{a}$ and $-\frac{\vec{b}}{2}$ will be 150° .

Reason :- Sign of dot product of two vectors tells you whether angle between two vectors is acute or obtuse

135. Assertion :- Vector product of two vectors may be greater than, equal to or less than the scalar product.

Reason :- At $\theta = 45^\circ$, two are equal.

136. Assertion :- Angle between $\hat{i} + \hat{j}$ and $\hat{i}$ is 45° .

Reason :- $\hat{i} + \hat{j}$ is equally inclined to both $\hat{i}$ and $\hat{j}$ and the angle between $\hat{i}$ and $\hat{j}$ is 90° .

137. Assertion :- Component of $\vec{A}$ along $\vec{B}$ is not equal to component of $\vec{B}$ along $\vec{A}$.

Reason :- Component of $\vec{A}$ along $\vec{B}$ is $A \cos \theta$, and component of $\vec{B}$ along $\vec{A}$ is $B \cos \theta$, where θ is the angle between two vectors

138. Assertion :- Small displacement is a vector quantity.

Reason :- Mass and distance are also vector quantities

139. Assertion :- The sum of squares of cosines of angles made by a vector with X, Y and Z-axis is equal to unity.

Reason :- A vector making 45° with x-axis must have equal components along X and Y - axis

140. Assertion :- Adding a scalar to a vector of the same dimensions is a meaningful algebraic operation.

Reason :- Displacement can be added to distance.

141. Assertion : The electronic configuration of nitrogen atom is represented as:



and not as :



Reason : The electronic configuration of the ground state of an atom is the one which has the greatest multiplicity.

142. Assertion : An orbital cannot have more than two electrons.

Reason : The two electrons in an orbital create opposite magnetic field.

SPACE FOR ROUGH WORK

- | | |
|---|---|
| <p>143. Assertion : A spectral line will be seen for $2p_x-2p_y$ transition.
 Reason : Energy is released in the form of wave of light when the electron drops from $2p_x$ to $2p_y$ orbital.</p> <p>144. Assertion : In Rutherford's gold foil experiment, every few α -particles are deflected back.
 Reason : Nucleus present inside the atom is heavy</p> <p>145. Assertion : There are two spherical nodes in 3s-orbital.
 Reason : There is no planar node in 3s-orbital.</p> <p>146. Assertion : If the potential difference applied to an electron is made 4 times, the de-Broglie wavelength associated is halved.
 Reason : On making potential difference 4 times, velocity is doubled and hence λ is halved.</p> <p>147. Assertion : On increasing the intensity of incident radiation, the number of photoelectrons ejected and their KE increases.
 Reason : Greater the intensity means greater the energy which in turn means greater the frequency of the radiation.</p> <p>148. Assertion : In third energy level there are three subshells.
 Reason : For $n = 3$, l can have three values 0, 1 and 2.</p> | <p>149. Assertion: Emitted radiation will fall in visible range when an electron jump from $n = 4$ to $n = 2$ in H atom.
 Reason: Balmer series radiation belongs to visible range for hydrogen atom.</p> <p>150. Assertion : The e/m of cathods rays does not depend upon nature of gas.
 Reason : Cathode rays are stream of electron which is a fundamental particle of atom.</p> <p>151. Assertion : The charge to mass ratio of the particles in anode rays depends on nature of the gas taken in the discharge tube.
 Reason : The particles in anode rays carry positive charge.</p> <p>152. Assertion : s-orbital cannot accommodate more than two electrons.
 Reason : s-orbitals are spherically symmetrical</p> <p>153. Assertion : The existence of three unpaired electrons in phosphorous atom can be explained on the basis of Hund's rule.
 Reason : According to Hund's rule, the degenerate orbitals are singly occupied and only then pairing takes place</p> <p>154. Assertion : If an electron is located within the range of $0.1 \text{ \AA}$ then the uncertainty in velocity is approximately $6 \times 10^6 \text{ m/s}$.
 Reason : Trajectory (path of motion) of above electron can be defined. [$h = 6.6 \times 10^{-34}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$]</p> |
|---|---|

SPACE FOR ROUGH WORK

- | | |
|--|---|
| <p>155. Assertion : Hydrogen has one electron in its orbit but it produces several spectral lines.
 Reason : There are many excited energy levels available.</p> <p>156. Assertion : The 19th electron in potassium atom enters into 4s-orbital and not the 3d-orbital.
 Reason : $(n + 1)$ rule is followed for determining the orbital of the lowest energy state.</p> <p>157. Assertion : For $n = 3$, $l = 0, 1$ and 2 and m may be $0; 0, \pm 1$ and $0, \pm 1$ and ± 2.
 Reason : For each value of n, there are 0 to $(n - 1)$ possible values of l and for each value of l, there are 0 to $\pm l$ values of m.</p> <p>158. Assertion : $2p_x, 2p_y$ and $2p_z$ each have one nodal plane.
 Reason : These orbital are degenerate orbitals.</p> <p>159. Assertion : Wave length of first line of any series in hydrogen spectrum is biggest among all other lines of the same series.
 Reason : Wavelength of a spectral line for an electronic transition is inversely related to difference in the energy levels involved in the transition.</p> <p>160. Assertion : Bohr's orbits are also called stationary states.
 Reason : Energies of Bohr's orbits are fixed.</p> <p>161. Assertion : Cell is an open system.
 Reason : Cell receives a number of materials including energy containing nutrients from outside.</p> | <p>162. Assertion : Interphase nucleus posses a number of short thick rod like deeply stained bodies.
 Reason : Chromatin contain DNA, RNA, histone proteins but non histone are absent.</p> <p>163. Assertion : Cilia work like oars and are smaller than flagella.
 Reason : The core of cilia or flagella is called axoneme.</p> <p>164. Assertion : Golgi apparatus remains in close association with the ER.
 Reason : Materials to be packaged in the form of vesicles from the ER fuse with the cis face of golgi body and move towards the trans face.</p> <p>165. Assertion : Chromosome are divided into two parts - euchromatin and heterochromatin.
 Reason : Heterochromatin are those regions that remain condensed during Interphase and rest of the non-condensed chromosomes are called euchromatin.</p> <p>166. Assertion : The ribosomes of chloroplast are smaller than cytoplasmic ribosomes.
 Reason : Chloroplast is semi-autonomous organelle.</p> <p>167. Assertion : Cells have the ability to adopt a variety of shapes and carry out directed movements.
 Reason : There are two principal types of protein filaments-actin filament and intermediate filaments, which constitute the cytoskeleton.</p> |
|--|---|

SPACE FOR ROUGH WORK

- | | |
|--|--|
| <p>168. Assertion : ER divides intracellular space into two distinct compartments.
 Reason : Endoplasmic reticulum is a network or reticulum of tiny tubular structures scattered in cytoplasm.</p> <p>169. Assertion : Facilitated diffusion is a type passive transport which occur through the carrier proteins.
 Reason : Since polar molecules can't cross the lipid bilayer they need carrier protein for transport.</p> <p>170. Assertion : Usually plant cells have less number of mitochondria than animal cells
 Reason : The number of mitochondria is related to the physiological activity of cell.</p> <p>171. Assertion : Biomolecules which are found in the acid insoluble fraction are consider as macromolecules.
 Reason : Biomolecules found in acid inoluble fraction always have molecular weight less than 800 Dalton</p> <p>172. Assertion : Lipid whose molecular weight does not exceed 800 Dalton come under acid insoluble fraction
 Reason : Lipid are present not only as such but also arranged in structures like membranes during grinding these lipid from vesicle.</p> | <p>173. Assertion : Lipids are strictly macromolecules
 Reason : Molecular weight of lipids exceeds more than 800 Dalton</p> <p>174. Assertion : Starch can be stained with iodine
 Reason : Cellulose can be stained with iodine</p> <p>175. Assertion : Glucose is a reducing sugar
 Reason : Glucose contain free ketone group</p> <p>176. Assertion : Glucose and mannose are epimers
 Reason : Isomers formed as a result of interchange of the –OH and –H groups on carbon atom 2,3 and 4 of glucose are called epimers</p> <p>177. Assertion : Sucrose is also called as invert sugar.
 Reason : Aqueous solution of fructose is laevorotatory.</p> <p>178. Assertion : Malt sugar is an example of diaccharide.
 Reason : Maltose is made up of α -D-glucose and α -D-glucose</p> <p>179. Assertion : Chitin is also called fungal cellulose
 Reason : Chitin is a linear polymer of N-acetyl-D-glucosamine</p> <p>180. Assertion : Chlorophylls are primary metabolites of plants
 Reason : Chlorophylls are required for photosynthesis which is a basic metabolic process of plants</p> |
|--|--|

SPACE FOR ROUGH WORK